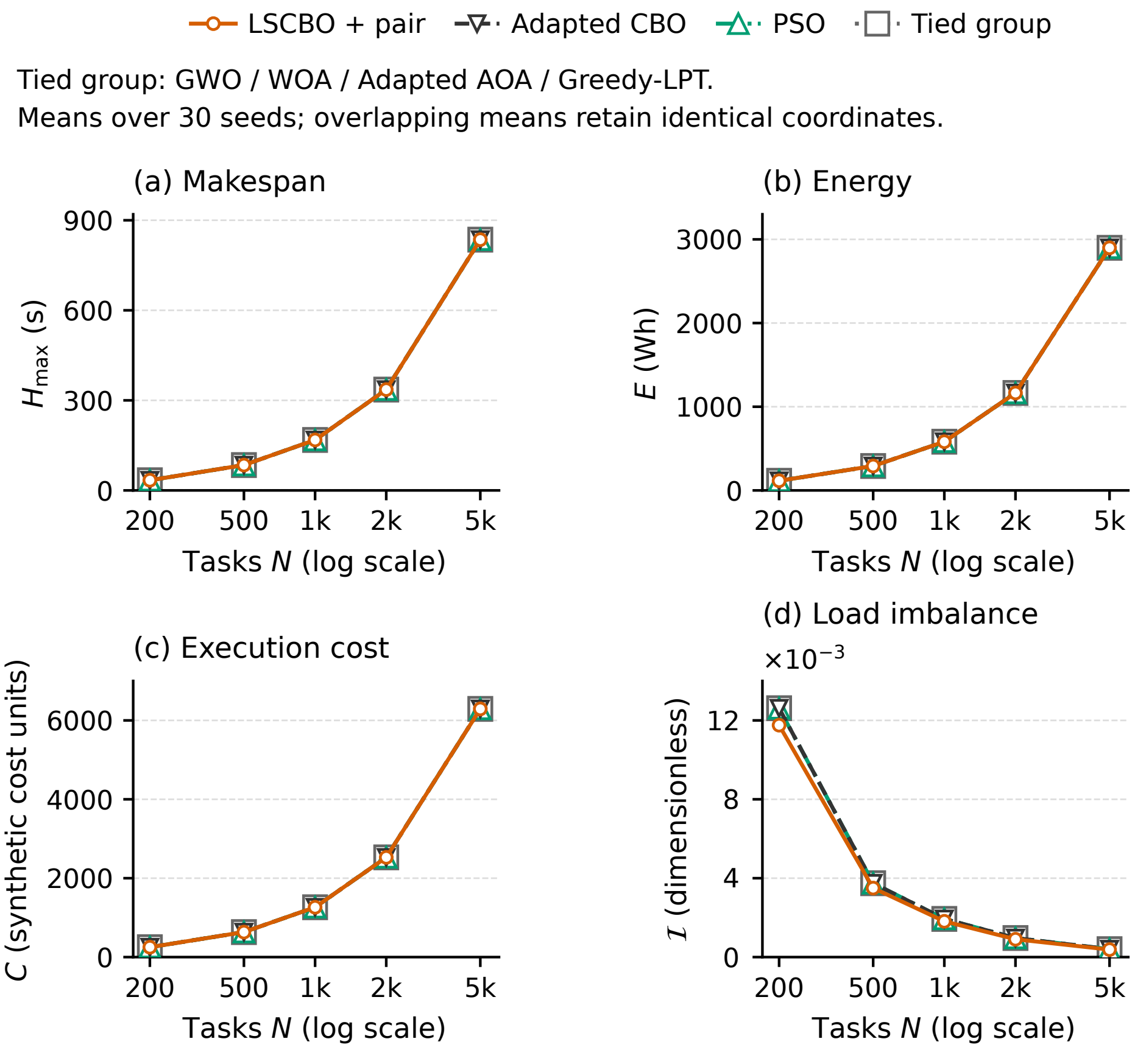
LSCBO + pair
Adapted CBO
PSO
Tied group
Tied group: GWO / WOA / Adapted AOA / Greedy-LPT.
Means over 30 seeds; overlapping means retain identical coordinates.
(a) Makespan
H_{max} (s)
900
600
300
0
200
500
1k
2k
5k
Tasks N (log scale)
(b) Energy
E (Wh)
3000
2000
1000
0
200
500
1k
2k
5k
Tasks N (log scale)
(c) Execution cost
C (synthetic cost units)
6000
4000
2000
0
200
500
1k
2k
5k
Tasks N (log scale)
(d) Load imbalance
$\times 10^{-3}$
$\mathcal{I}$ (dimensionless)
12
8
4
0
200
500
1k
2k
5k
Tasks N (log scale)